Δ vs frozen (good-oriented, value + 95% CI) · PRETRAIN block / SURGERY block · SCORE = mean per-metric-normalized (0–1)
WINNER row (champion duel — S=surgery, P=pretrain, ==tie): surgery 4 · pretrain 1 · tie 4 → SURGERY

	action_top1 ↑ better classify motion class from frozen features	motion_cos ↑ better same-motion clips cluster vs different motions	taxonomy_f1 ↑ better read scene attributes from features	future_mse ↓ better predictor guesses masked next-frame tokens	rollout ↓ better error growth over multi-step rollout	causal ↓ better predict 2nd half from 1st half	tdist ↓ better accuracy decay vs prediction horizon	teacher_free ↓ better free-run minus teacher-forced gap	maskratio ↓ better degradation under heavier masking	SCORE (norm 0–1)
pretrain_2X encoder	+8.77 [+6.466, +11.014]	+0.160 [+0.156, +0.164]	-0.0077 [-0.024, +0.008]	+0.011 [+0.010, +0.012]	-0.0003 [-0.000, -0.000]	+0.012 [+0.011, +0.012]	-0.0013 [-0.001, -0.001]	+0.0019 [+0.002, +0.002]	+0.0008 [+0.001, +0.001]	0.44
pretrain encoder	+7.29 [+4.986, +9.479]	+0.150 [+0.146, +0.154]	-0.0028 [-0.018, +0.013]	+0.017 [+0.017, +0.018]	+0.0002 [+0.000, +0.000]	+0.019 [+0.018, +0.019]	+0.0001 [-0.000, +0.000]	+0.0028 [+0.003, +0.003]	+0.0036 [+0.003, +0.004]	0.74
pretrain head	-1.53 [-3.652, +0.493]	+0.0036 [+0.003, +0.004]	-0.0005 [-0.014, +0.013]	-0.0006 [-0.001, -0.000]	0 [+nan, +nan]	0 [+nan, +nan]	0 [+nan, +nan]	0 [+nan, +nan]	0 [+nan, +nan]	0.35
surgery 3stage_DI encoder	+5.92 [+3.616, +8.219]	+0.154 [+0.150, +0.159]	-0.0006 [-0.016, +0.015]	+0.042 [+0.041, +0.043]	-0.0006 [-0.001, -0.000]	+0.037 [+0.036, +0.037]	-0.0009 [-0.001, -0.001]	-0.0022 [-0.003, -0.002]	+0.013 [+0.012, +0.013]	0.67
surgery 3stage_DI head	+6.85 [+4.548, +9.151]	+0.158 [+0.154, +0.163]	-0.0044 [-0.020, +0.011]	+0.018 [+0.017, +0.018]	+0.0002 [+0.000, +0.000]	+0.019 [+0.018, +0.019]	+0.0001 [-0.000, +0.000]	+0.0028 [+0.003, +0.003]	+0.0036 [+0.003, +0.004]	0.71
surgery noDI encoder	+7.40 [+4.931, +9.644]	+0.159 [+0.155, +0.163]	-0.0034 [-0.020, +0.013]	+0.042 [+0.041, +0.043]	-0.0005 [-0.001, -0.000]	+0.038 [+0.037, +0.038]	-0.0015 [-0.002, -0.001]	-0.0009 [-0.001, -0.001]	+0.010 [+0.010, +0.011]	0.61
surgery noDI head	+7.34 [+4.986, +9.644]	+0.170 [+0.166, +0.175]	-0.0051 [-0.021, +0.011]	+0.017 [+0.016, +0.018]	+0.0002 [+0.000, +0.000]	+0.019 [+0.018, +0.019]	+0.0001 [-0.000, +0.000]	+0.0028 [+0.003, +0.003]	+0.0036 [+0.003, +0.004]	0.71
WINNER	P	S	=	S	=	S	=	=	S	